

AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
538 A	30.80 MVA	90.0 degC	78.4 degC	29.875 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		240	mm2
Conductor type		stranded	
Conductor diameter Dc		18.4	mm
Conductor screen OD Dc_sc		19.4	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		34.8	mm
Ins outer screen OD Di_sc		35.8	mm
Insulation thickness t_ins		7.70	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
Screen type		Cu_wire	
Screen mean diameter ds		36.8	mm
Screen area As		35.6	mm2
Armour type		none	
Oversheath material		PE	
Cable outer diameter De		44.0	mm
Rated voltage U		33.0	kV
U0 line-to-earth		19.053	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		44.0	mm
Ambient ground temp theta_a		20	degC
Soil thermal resistivity rho_soil		1.00	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		7.5400e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 20 degC		7.5400e-05	ohm/m
R0 at theta_max = 90 degC		9.6143e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
$xs4 = [ks(8\pi i f / R0) \times 10^{-7}]^2$		1.708396	
ys skin effect factor		0.008835	
$xp4 = [kp(8\pi i f / R0) \times 10^{-7}]^2$		1.093373	
yp" intermediate proximity		0.005669	
yp proximity factor		0.004298	
R at theta_max AC = $R0(1+ys+yp)$		9.7405e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
U0 line-to-earth		19.0526	kV
Capacitance C = $er / (18 \ln(Di_sc / Dc_sc)) \times 10^{-9}$		0.226693	nF/m
$\omega = 2 \pi f$		314.1593	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		1.0341e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.63	mm2
Lay factor LFs		1.115139	
Screen Rs0 at 20 degC (with LFs)		5.3960e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_sc semiconducting screen		2.50	K.m/W
rho_tape bedding tape		6.00	K.m/W
T1_cs conductor screen		0.021057	K.m/W
T1_i insulation		0.325505	K.m/W
T1_si insulation screen		0.011272	K.m/W
T1_uwbt bedding tape under wire screen		0.026308	K.m/W
T1_raw = sum of layers		0.384142	K.m/W
T1 correction factor x1.07 (wire coverage check)		0.411032	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
T3_raw oversheath		0.062913	K.m/W
T3 x1.6 touching trefoil factor		0.100660	K.m/W
u = 2L/De		45.4545	
T4 soil		1.852497	K.m/W
Sum T = T1+T2+T3+T4		2.364189	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		78.3503	degC
Rs at theta_s		6.6334e-04	ohm/m
X sheath reactance		5.3261e-05	ohm/m
lam1' circulating current		0.043622	
lam1" eddy current		0.008714	
lam1 = lam1' + lam1"		0.052336	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		70.00	K
Etop numerator		69.776775	
Ebot denominator		2.4024e-04	
I = sqrt(Etop/Ebot) exact		538.9295	A
I rounded TB880 GP2		538	A
MVA = sqrt(3) x U x I		30.804	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		78.3503	degC
theta_j cable outer surface		75.3431	degC
theta_a ambient ground		20.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I ² x R		28.290835	W/m
Sheath Ws = lam1 x Wc		1.480625	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		1.0341e-01	W/m
Total		29.874868	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.174368	uH/m
Reactance X = omega L		54.7793	uOhm/m
Pos-seq resistance R1		102.5029	uOhm/m
Pos-seq impedance Z1		116.2223	uOhm/m
Capacitance C		0.226693	nF/m
Charging current Ic		1.3569	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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